

Biological Ageing Pace as a Predictor of Alzheimer's Conversion in MCI

*A Statistical and Machine Learning Analysis
Using DunedinPACNI and Cognitive Reserve*

AT A GLANCE

614

MCI participants

32.4%

3-year converters

OR 1.25

DunedinPACNI effect

OR 3.13

APOE ϵ 4 effect

AUC 0.87

Best ML model

The Clinical Problem

57M

people living with dementia (2021)

152M

projected by 2050

\$1.3 Trillion

global annual cost (2019)

New treatments — new problem

- Lecanemab & Donanemab approved 2023
- Risk: ARIA — cerebral oedema & microhaemorrhages
- Benefit: modest slowing (months, not years)
- Ethical use requires accurate risk stratification at MCI stage

MCI is profoundly heterogeneous

- ~10–15% progress to AD per year in clinical settings
- 32.4% converted over 3 years in this ADNI cohort
- Some remain stable; some revert to normal
- Single biomarkers leave individual-level prediction imperfect

What This Study Tested

01

Validate DunedinPACNI

Does biological ageing pace independently predict 3-year MCI-to-AD conversion after adjusting for age, sex, APOE ϵ 4?

02

Test Cognitive Reserve

Does years of formal education independently predict conversion and improve model fit beyond PACNI + covariates?

03

Test the Interaction

Does education moderate the DunedinPACNI–conversion relationship? (Diathesis-stress / buffer hypothesis)

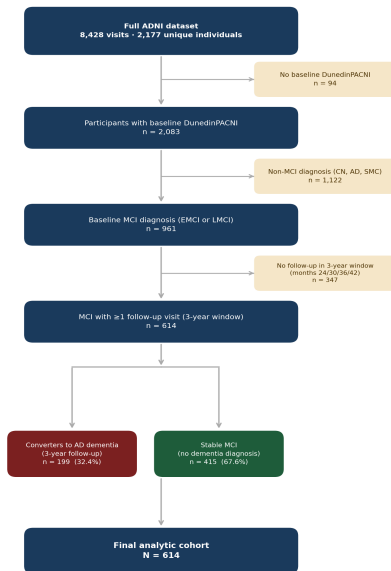
04

Machine Learning Benchmark

Do 5 classifiers under nested 5-fold CV achieve clinically meaningful discriminative performance?

Data & Analytical Design

Participant Selection Flow



No missing data in any key analytic variable (DueneidPACNI, education, age, sex, APOE4), VIF < 2.5 for all predictors. No influential outliers (Cook's D).

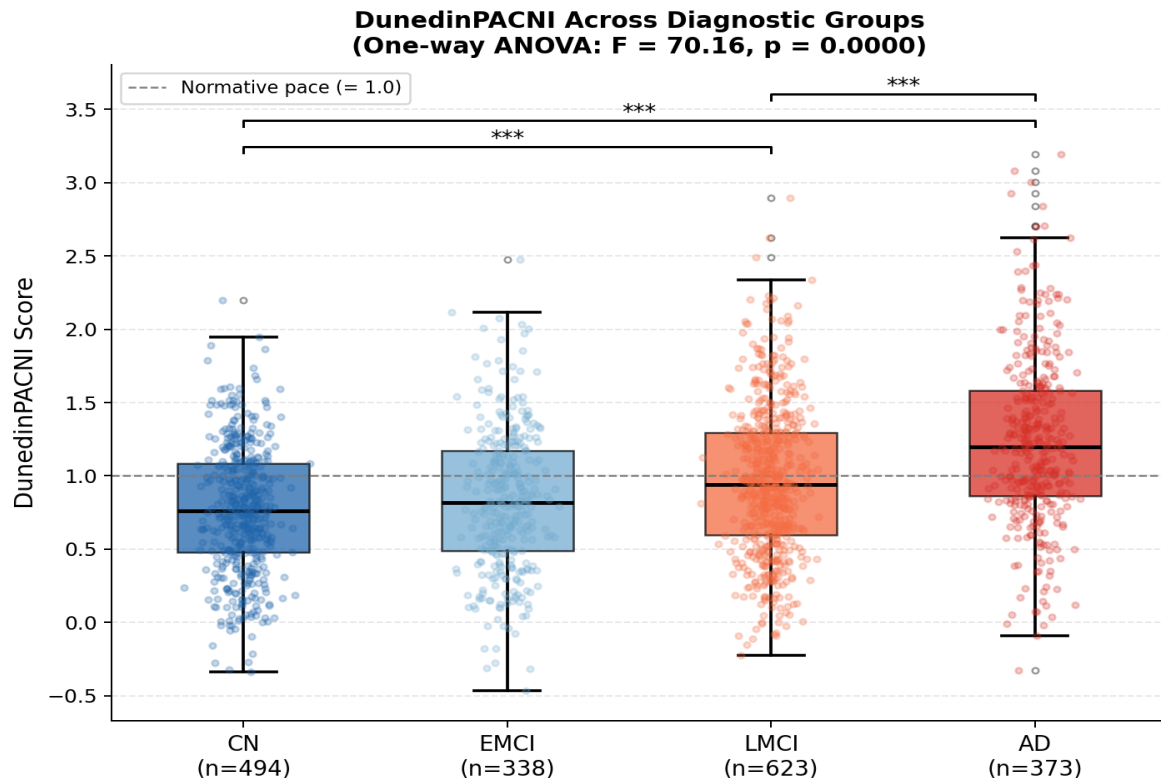
Statistical Analysis — R 4.3.1

- Three nested logistic regression models
- All predictors z-standardised (puts variables on the same scale and improves interpretation)
- VIF < 1.3 (Variance Inflation Factor; checks for multicollinearity between predictors)
- Hosmer–Lemeshow $\chi^2(8) = 10.24$, $p = .248$ (tests model calibration; non-significant result indicates good fit)
- AUC with DeLong confidence intervals (measures predictive discrimination and quantifies uncertainty)
- Supplementary cognition-adjusted analysis (tests whether PACNI predicts conversion beyond baseline cognition)
- CSF biomarker subsample analysis, $n = 339$ (tests whether PACNI remains informative after accounting for established Alzheimer's biomarkers)

Machine Learning — Python 3.11, scikit-learn 1.3, XGBoost 2.0

- Machine Learning Pipeline
- 5 classifiers: Logistic Regression, Random Forest, Support Vector Machine (SVM), K-Nearest Neighbours (KNN), and XGBoost
- True nested 5-fold cross-validation (hyperparameter tuning performed within each training fold; provides an unbiased estimate of performance)
- GridSearchCV in the inner loop (systematically searches for the optimal hyperparameter combination)
- Class imbalance handling: balanced class weights / scale_pos_weight = 2.09 (prevents the model favouring the larger stable-MCI group)
- Performance evaluated using AUC, calibration, confusion matrices, learning curves, and SHAP explainability analyses

DunedinPACNI Rises Across the AD Spectrum



$$F(3, 1824) = 70.16$$
$$p < .001$$

One-way ANOVA

Cohen's $d = 0.95$

- CN vs AD — large effect size
- CN–EMCI non-significant

What this raises

- PACNI tracks disease severity across the continuum
- Key question: ageing biomarker or disease severity proxy?

Logistic Regression: All Three Models

Variable	Model 1	Model 2	Model 3
DunedinPACNI	1.25 (1.03–1.51)*	1.25 (1.04–1.52)*	1.24 (1.03–1.51)*
Education	—	0.91 (0.76–1.08)	0.90 (0.75–1.07)
PACNI × Education	—	—	1.07 (0.90–1.29)
APOE ε4 carrier	3.13 (2.18–4.54)***	3.12 (2.17–4.51)***	3.11 (2.17–4.51)***
AUC (95% CI)	0.668 (0.622–0.714)	0.671 (0.625–0.716)	0.673 (0.628–0.718)

DunedinPACNI

- Significant across all 3 models
- OR 1.24–1.25, all $p \leq .025$
- 25% ↑ odds per SD

APOE ε4 dominates

- OR 3.11–3.13 across all models
- Strongest predictor
- Non-redundant with PACNI

Education: null

- OR 0.91, $p = .274$
- Interaction OR 1.07, $p = .441$
- Inconclusive — not definitively negative

When Baseline Cognition Enters the Model

Without Cognition
(Primary Model)

OR = 1.25

95% CI: 1.03–1.51 $p = .022$ ✓



+ cognition

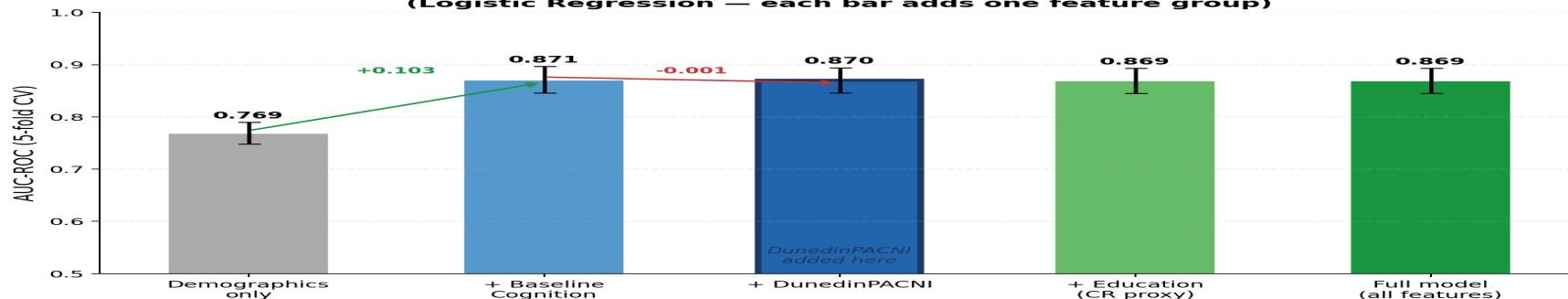
With Cognition
(SA1 Supplementary)

OR = 0.98

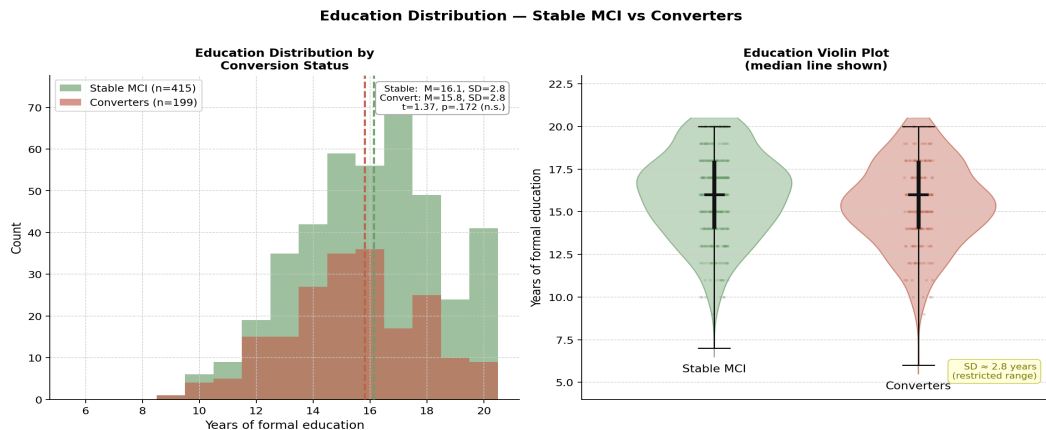
95% CI: 0.78–1.23 $p = .882$ ✗

Cognition
OR 4.84

Incremental Feature Contribution to MCI-to-AD Conversion Prediction
(Logistic Regression — each bar adds one feature group)



The Education Null — Not What It Seems



ADNI: $M = 16.0$ yrs, $SD = 2.8$ — effective range only 12–20 yrs
Population $SD \approx 4\text{--}5$ yrs $\rightarrow$ power reduced $\sim 60\text{--}70\%$ (McClelland & Judd, 1993)

1 Restriction of range

SD = 2.8 vs population $\sim 4\text{--}5$ yrs $\rightarrow$ 60–70% power loss

2 Proxy inadequacy

Education conflates quality, timing, neural enrichment

3 Reserve depletion at MCI

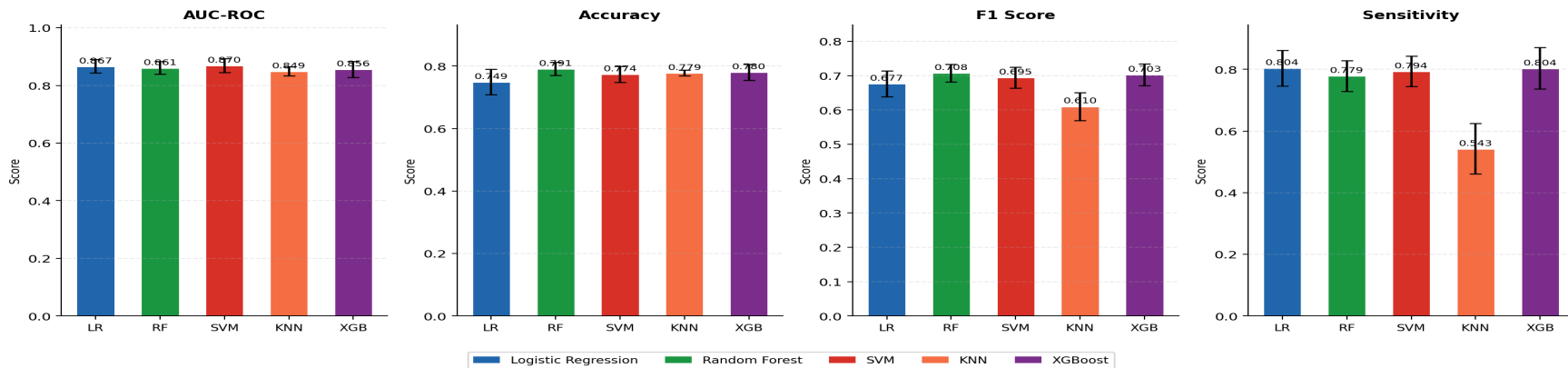
High-reserve individuals may have exhausted compensatory capacity

4 Genuine null (possible)

Within a 3-year window, protection may simply not manifest

ML Performance & What Drives It

Model Performance Comparison (5-Fold Nested CV, mean $\pm$ SD)



All 5 classifiers

- AUC 0.849–0.870 under nested CV
- SVM highest AUC (0.870)
- LR matches XGBoost at 0.867
- Complexity adds nothing

Ablation tells the story

- Demographics only: AUC 0.769
- + Baseline cognition: $\rightarrow$ 0.871 (+0.103)
- + DunedinPACNI: $\rightarrow$ 0.870 (−0.001)
- + Education: $\rightarrow$ 0.869 (−0.001)

Key implication

- All performance from cognitive assessment
- PACNI marginal gain $\approx$ zero once cognition known
- Consistent with SA1 attenuation finding

Four Key Contributions

1

DunedinPACNI predicts conversion

OR = 1.25, $p = .022$ — stable across 3 models, independent of APOE $\epsilon 4$. 25% $\uparrow$ odds per SD. Replicates Whitman et al. 2025 in a different framework.

2

Constrained by baseline disease severity

OR attenuates to 0.98 ($p = .882$) when cognition added. Confirmed by ML ablation ($\Delta\text{AUC} = -0.001$). Limited incremental value in established MCI.

3

Education null = informative, not definitive

ADNI's high-attainment ceiling (SD = 2.8 yrs) reduces power by ~60–70%. The moderation hypothesis remains untested, not refuted.

4

ML performance driven by cognition, not PACNI

AUC 0.87 from 7-feature models vs 0.67 primary. Gain overwhelmingly from cognitive composite. Simple LR matches XGBoost.

Where Next?

Limitations

ADNI: highly selected, predominantly White, highly educated volunteer sample

DunedinPACNI trained on European-ancestry NZ cohort — cross-ethnic validity unconfirmed

Binary 3-year outcome: misclassification risk; time-to-event would be better

Education is a poor reserve proxy in a high-attainment cohort

Cross-sectional baseline: cannot distinguish mediation from overlap

Amyloid/tau PET unavailable for most participants



Future Directions

CRITICAL

Pre-MCI longitudinal cohort measuring PACNI serially before symptoms — distinguishes mediation vs overlap

HIGH

Composite reserve indices: occupational complexity + leisure + social connectivity

HIGH

Educationally diverse, ethnically diverse cohorts — properly power the moderation test

MEDIUM

Formal mediation analysis with time-varying PACNI and cognition as mediator

MEDIUM

Time-to-event Cox framework for greater statistical power

Thank You

Questions & Discussion

DunedinPACNI is a biologically meaningful, non-invasive marker of vulnerability in MCI — with most clinical value before MCI onset.

KEY NUMBERS

N = 614

final MCI cohort

OR 1.25*

DunedinPACNI

OR 3.13***

APOE ϵ 4

p = .882

after cognition adj.

AUC 0.87

best ML classifier

Δ AUC -0.001

PACNI marginal gain